MILITARY SPECIFICATION

WET SUITS, NEOPRENE

1. SCOPE

1.1 Scope. - This specification covers fire resistant sponge neoprene five (5) piece wet suits including jacket, trousers, hood, boots and one-finger mitts for use by shipboard swimmers and lifeboat crews for rescue work.

1.2 Classification.-

1.2.1 Types. - The wet suits covered by this specification shall be the type used by shipboard swimmers and lifeboat crews for rescue work.

Type I - 3/16 inch neoprene

Type II - 1/4 inch neoprene

1.2.2. Sizes.- The jacket and trousers of the wet suit shall be of the following sizes: (See 3.3)

Small, Medium, Large and Extra Large.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards and publications of the issue in effect on the date of invitation for bids form a part of this specification:

SPECIFICATIONS

Federal

V-F-106 - Fasteners, Slide, Interlocking

PPP-B-636 - Box, Fiberboard

STANDARDS

Federal

FED-STD-751 - Stitches, Seams and Stitching

Military

MIL-STD-129 - Marking for Shipment and Storage

FSC 4220

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OTHER PUBLICATIONS

American Society for Testing and Material Standards.

D-412-62T "Tension Testing of Vulcanized Rubber"

D-1056-62T "Tentative Specifications and Methods of Test for Sponge and Expanded Cellular Rubber Products"

D-1149-62T "Tentative Method of Test for Accelerated Ozone Cracking of Vulcanized Rubber"

3. REQUIREMENTS

3.1 Preproduction Sample. - When stipulated in the procurement documents and as soon as practicable after award of the contract or order and prior to submission of any completed units for final acceptance, the contractor shall furnish a production sample of the wet suit complete with jacket, trousers, hood, boots, and one-finger mitts for preproduction inspection and testing to determine the conformance with the specification. A manufacturer's test report on the neoprene material used in the production of the suit, composite of nylon backed neoprene and seam construction, shall also be submitted at this time. The test report shall be from a testing laboratory acceptable to the government and shall show that the neoprene materials and construction conforms to the requirements of paragraphs 3.2.1 and 3.5.2. Approval of the preproduction sample by the activity concerned shall not relieve the contractor of his obligation to supply equipment conforming to this specification. No changes or deviation from the preproduction sample will be acceptable without prior written approval from the contracting officer.

3.2 Materials. -

3.2.1 Neoprene. - The neoprene shall be either gas blown or chemical stock, Type "S" cellular rubber of ASTM Standard D-105b-62T having a closed cell structure and shall be formulated so that it will not support combustion The neoprene stock used may be either two skin or one skin (slit sheet), and shall conform to the following:

3.2.1.1 Compression Deflection. -The neoprene stock shall have a compression deflection of 2 to 5 psi to deflect 25 percent when tested as specified in 4.4.1.1.

3.2.1.2 Water Absorption. - The water absorption of the neoprene stock shall be no greater than 5 percent by weight when tested as specified in 4.4.1.2.

3.2.1.3 Tensile Strength. - Both Type I and Type II nylon backed neoprene stock shall have a direct pull tensile strength for a 1" x 6" specimen greater than 25 pounds when tested as specified in 4.4.1.3.

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3.2.1.4 Modulus. - The direct pull modulus for a 1" x 6" specimen of the nylon backed neoprene stock at 100 percent elongation shall be 4 to 10 pounds for Type I and 5 to 12 pounds for Type II when tested as specified in 4.4.1.4.

3.2.1.5 Elongation at Break. - When tested as specified in 4.4.1.5 the elongation at break shall be 200 percent or greater for both Type I and II nylon backed neoprene stock.

3.2.1.6 Accelerated Aging. - When tested as specified in 4.4.1.6, the change from the original load deflection shall not exceed ± 20 percent.

3.2.1.7 Ozone Resistance. - The coated neoprene stock shall resist ozone attack at 50 pphm by volume concentration for at least 7 days at 104°F when tested as specified in 4.4.1.7.

3.2.1.8 Density. -The density of the neoprene stock shall be from 10 to 20 pounds per cubic foot.

3.2.2 Nylon Lining. - The neoprene stock shall be reinforced by bonding a knitted nylon stretch cloth in the inner side of the neoprene stock. When slit sheet stock is used, the reinforceing material shall be bonded to the exposed cell side of the sheet. The knit material shall be made of 100 percent continous filament stretch-type nylon yarn. The material shall conform to the physical requirements listed in Table I.

TABLE I

Stitch type	Jersey
Weight, ounces per square yard	2.0 to 4.0 ounces
Wales per inch (minimum)	20
Courses per inch (minimum)	30
Elongation, wales (percent)	150 minimum
Elongation, courses (percent)	200 minimum
Color	black

Note. - The cloth shall be scored, dyed and heat set, and when finished shall show good colorfastness to laundering, crocking and perspiration.

3.2.3 Zippers and Fasteners. - All zippers shall have nickel silver chains with water and mildew resistant tapes in accordance with Specification V-F-106. The zippers used on the wrists of the jacket and ankles of the trousers shall be 8 inches long, have long tab pulls and conform to Type I, style 3, size M or MS of Specification V-F-106. The zippers for the jacket fronts shall be full length, have wire stirrup pulls, and conform to Type IV, Style 7, Size M or MS of Specification V-F-106. The fasteners on the bottom of the jackets for the beaver tail piece shall be the "common sense" ("turn button") type and shall be of nickel plated brass.

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3.2.4 Coating.- The outer surface of the neoprene stock (skinned surface for slit sheet) shall be coated with a protective coating which is compatible with the neoprene stock. The color of the coating shall be orange matching Federal Standard No. 595, color 12197. The coating shall have good adhesion to the neoprene, and shall have an elasticity equivalent to the neoprene stock. The coating shall not support combustion and shall be non-embrittling.

3.3 Sizing. - The jacket and trousers of the suits shall be form fitting, and shall conform to the body measurements listed in Table II.

TABLE II

SIZE CHART

	Small	Medium	Large	Extra Large
Chest	35" - 37"	37" - 40"	40" - 42"	42" - 45"
Waist	29" - 33"	32" - 35"	34" - 37"	37" - 41"
Weight	130 - 155 lbs.	150 - 180 lbs.	170 - 200 lbs.	200 - 235 lbs.
Height	5'4" to 5'8"	5'7" to 6'0"	5'10" to 6'2"	6'0" to 6'4"

3.4 Design. -

3.4.1 Jacket. - The jacket design shall be full cut, natural shoulders of five piece construction exclusive of zipper protective flaps and reinforcing pads. The jacket shall follow the pattern pieces shown in Drawing I. These patterns do not represent exact dimensions and shall be adjusted to conform to the body measurements of Table II. The jacket shall be fitted with a front zipper and wrist zippers. The front zipper shall terminate at the normal collar line to provide personnel comfort. All zippers shall be fitted with zipper protective flaps (see 3.5.4).

3.4.2 Trousers. - The trousers shall have a high rise waist and be of four piece construction exclusive of zipper protective flaps and reinforcing pads. The trousers shall have crotch reinforcement and follow the pattern pieces shown in Drawing 2. These patterns do not represent exact dimensions and shall be adjusted to conform to the body measurements of Table II. The trousers shall be fitted with ankle zippers on the outside of the ankle and with zipper protective flaps (see 3.5.4).

3.4.3 Hoods. - The hoods shall be form fitting with contoured ears. The hoods shall be fabricated of neoprene stock with a cutout to expose the eyes, nose and mouth. The ear area of the hood shall contain a nest of 1/4 inch perforations consisting of a center perforation surrounded by a circle of six perforations, at a radius of 3/4 inch from the center perforation. The intent of these perforations is to facilitate hearing when the suits are used for rescue work. The hoods shall be sized to be compatible with the associated suit size.

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3.4.4 Boots. - The boots shall be molded neoprene foam stock with high top and semi-firm bottom. The boots shall be sized to be compatible with the associated suit size.

3.4.5 One Finger Mitts. - The mitts shall be fabricated of neoprene stock and shall be of one finger mitten design. The mitt shall extend one inch past the top of the zipper in jacket arm.

3.5 Construction.

3.5.1 Lamination of the Nylon Backing. - A nylon stretch fabric lining conforming to the requirements of paragraph 3.2.2 shall be smoothly and completely bonded to the neoprene stock used in the construction of the hood jacket and trousers of the suit. The lining shall be vulcanized or adhesively bonded using material compatible with the neoprene stock and the nylon lining. If slit sheet neoprene is used, the lining shall be bonded to the exposed cell side of the sheet.

3.5.2 Seams. - All seams shall be pressure joined using an appropriate adhesive. The seams of the jacket and trousers shall be reinforced so that seam strength is at least 30% that of the nylon backed sheet when tested as specified in 4.4.1.3. Seam strength shall be tested with the axis of pull perpendicular to the seam line. The modulus of the seam shall be the same as that of the nylon backed sheet along the seam line.

3.5.3 Attachment of Zippers and Fasteners. - The zippers shall be installed by adhesively bonding the zipper tapes to the inner surface of the suit. The attachment shall be further reinforced by stitching the tape with a double row of stitching using nylon thread. The stitches shall be spaced 3/16 to 1/4 inches apart and shall conform to stitch type 301 of standard FED-STD-751. The "common sense" or "turn button" fasteners shall be installed with bonded reinforcement pieces to prevent them from being pulled or torn from the neoprene stock.

3.5.4 Protective Zipper Flaps.- All zippers shall be fitted with protective zipper flaps to prevent the zipper catching or pinching the skin. The protective flaps shall be adhesively bonded to the material along the zipper line and adjacent to the zipper tape along one side of the zipper so that the flap extends beneath the zipper to protect the body from the zipper. The faying surface of the protective flap shall be tapered to provide as smooth an inner surface as possible. When slit sheet material is used for the flaps, the skinned surface shall face the interior of the suit.

3.5.5. Reinforcement Pieces.- The reinforcement pieces for the under arm seam junction and crotch seam shall be 1/8 inches thick neoprene stock with skinned surface facing the interior of the suit. The faying surface of the reinforceing pieces shall be tapered to provide as smooth an inner surface as possible. The reinforceing pieces shall be adhesively bonded in place.

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3.6 Hangers.- Plastic contoured hangers shall be provided for each suit. These hangers shall be fashioned so that they provide a large contact area with the jacket of the suit to prevent creases and tears in the shoulders of the jacket when on the hanger. The hangers shall have rust proof hooks to allow the trousers to hang without folding. The trousers shall be fitted with tabs to accomodate hanging.

3.7 Workmanship. - The components of the suit shall be uniform in quality and shall be free from irregularities or defects which could adversely affect the intended use of the suit.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of Tests and Inspections. - Tests and inspections shall be classified as follows:

a. Specimen tests
b. Production Sample inspection
c. Quality Conformance inspection

4.2.1 Specimen Tests.- Testing of material specimens (paragraph 3.2.1) shall be made on specimens of the neoprene stock to be used in the manufacture of a lot of suits. Where specified, the test shall be made on the nylon backed neoprene stock as used in the manufacture of the wet suits. Specimens for testing seam strength (paragraph 3.5.2) shall be prepared using seam fabrication techniques used in production. This testing includes the tests of 4.4.

4.2.2 Production Sample Inspection. - Examination and testing of production sample (paragraph 3.1) shall be made on a completely fabricated suit for all provisions of this specification applicable to the end product. This inspection shall include the examination of 4.3 and any of the tests of 4.4 deemed necessary to assure conformance with this specification.

4.2.3 Quality Conformance Inspection. - Quality conformance inspection shall be performed on all units offered under the contract. This inspection shall be the examination of 4.3.

4.3 Examination. - Each component of the suit shall be examined for design, construction, materials, operation of zippers and fasteners, and workmanship to determine compliance with the requirements of this specification. Any deviation from these requirements shall be cause for rejection of that component of the suit.

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4.4 Tests. -

4.4.1 The tests on the specimens of neoprene sheet and the nylon backed coat neoprene sheet shall be as follows:

4.4.1.1 Compression Deflection. - The neoprene sheet without nylon backing or coating used in the fabrication of the wet suits shall be tested for compression deflection by the method of ASTM-D-1056-62T paragraphs 20, 21, 22 and 23.

4.4.1.2 Water Absorption. - The neoprene sheet without nylon backing or coating used in the fabrication of the wet suit shall be tested for water absorption by the method of ASTM-D-1056-62T paragraphs 33, 34 and 35.

4.4.1.3 Tensile Strength. - The tensile strength of the nylon backed neoprene sheet shall be tested in accordance with ASTM-D-412-62T. The specimen size shall be 1 inch wide by 6 inches long and of the thickness used in fabrication. Tensile strength in this case is defined as the force applied at the time of rupture when the specimen is pulled along the 6 inch axis. The specimen for seam strength shall be cut from material from seamed nylon backed neoprene stock in which the seam was fabricated using production techniques. The seam shall be 3 inches from the end of the specimen and perpendicular to the line of pull.

4.4.1.4 Modulus. - The modulus of the nylon backed neoprene shall be tested in accordance with ASTM-D-412-62T. The specimen size to be 1 inch wide and 6 inches long and of the thickness used in fabrication. Modulus in this case is defined as the force applied resulting in 100 percent elongation. The specimen for testing seam modulus shall have the seam centered in the specimen parallel to the line of pull.

4.4.1.5 Elongation. - The elongation at break of the nylon backed neoprene shall be tested in accordance with ASTM-D-412-62T. The specimen shall be as in 4.4.1.3.

4.4.1.6 Accelerated Aging. - The accelerated aging of the neoprene stock shall be tested in accordance with ASTM-D-1056-62T paragraphs 18 and 19. Using air oven aging for 7 days at 158°F. The specimen shall be as required in 4.4.1.1. The results shall be reported as the percentage change of compression deflection values.

4.4.1.7 Ozone. - The coated neoprene stock shall be tested for resistance to ozone attack in accordance with ASTM-D-1149-62T using 50 pphm by volume ozone concentration for at least 7 days at 104°F.

5. PREPARATION FOR DELIVERY

5.1 Packaging. - The wet suits plus a contoured suit hanger for each suit, shall be lightly and neatly folded into a corrugated carton (11½ x 4½ x 28 inches) conforming to Type I, Class I, Style FTC or FOL of Specification PPP-B-636. The suits shall be packed in accordance with good commercial practices.

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5.2 Packing. - The wet suits with hangers packaged as specified shall be packed for shipment in standard commercial containers so constructed as to insure acceptance by common or other carrier for safe transportation, at the lowest rate, to the point of delivery. (Whenever practicable, suits of one size only shall be packed in the same box for shipment.) (Not less than 6 nor more than 12 packages shall be packed in one case for shipment.)

5.3 Marking. - Interior and exterior containers shall be marked in accordance with Standard MIL-STD-129. The nomenclature shall include the following:

Wet Suits, Neoprene
Specification MIL-W-82400(CG)
Manufacturer's Name
Contract Number
Date of Manufacture (month and year)

6. NOTES

6.1 Intended Use. - These suits are intended for use by shipboard swimmers and lifeboat crews engaged in rescue work. The type suit used will depend upon the severity of water and air temperature encountered.

6.2 Ordering Data. -

(a) Type required (see 1.2.1)
(b) Size required (see 1.2.2)
(c) Preproduction Sample (see 3.1)
(d) Title, number and date of this Specification

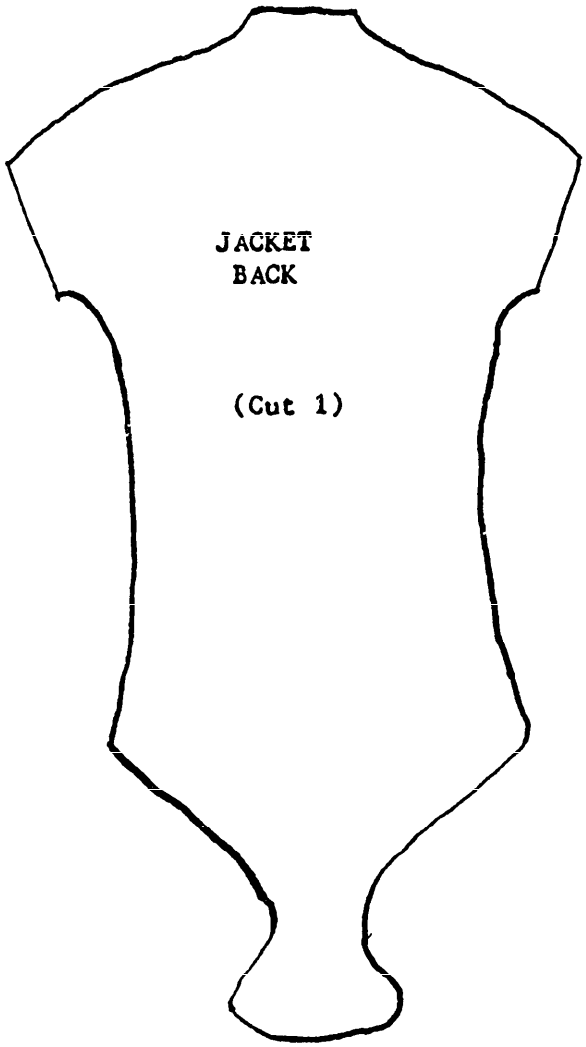
NOTICE. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

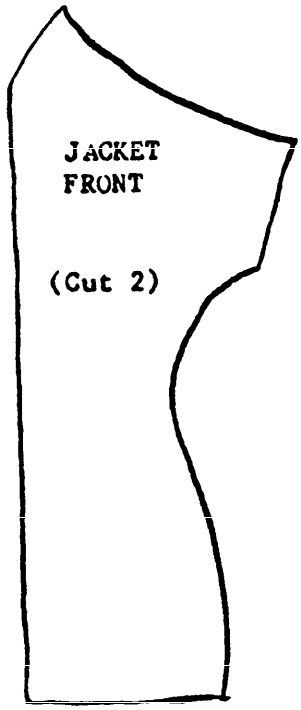
Preparing Activity:
U. S. Coast Guard

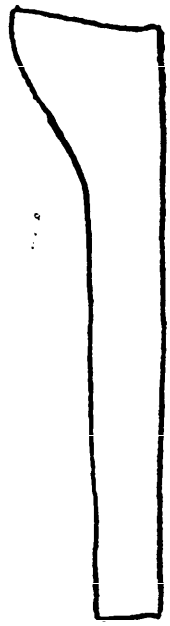
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DRAWING I

PATTERN PIECES, JACKET





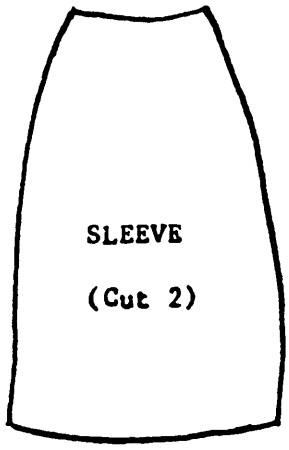


JACKET
PROTECTIVE FLAP
(zipper)

(Cut 1)

UNDERARM JUNCTION
REINFORCEMENT PIECE

(Cut 2)





SLEEVE
PROTECTIVE FLAP
(zipper)

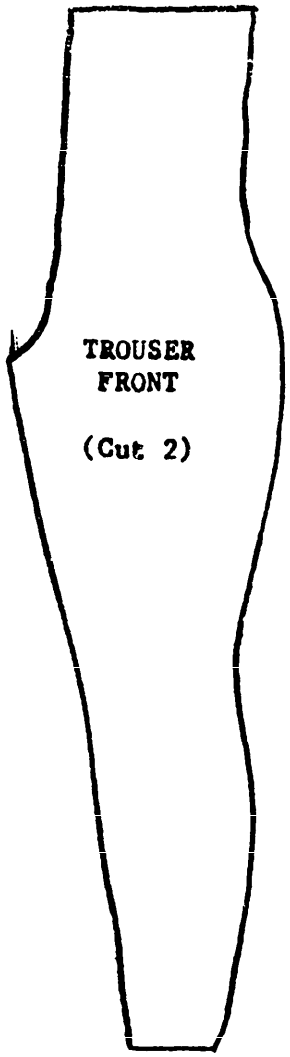
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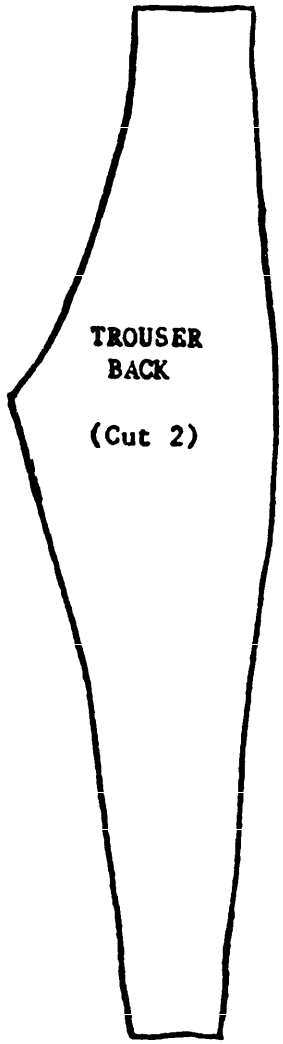
NOTE. - These pattern pieces do not represent exact measurements and are used for illustrative purposes only.

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DRAWING 11

PATTERN PIECES, TROUSERS







ANKLE
PROTECTIVE FLAP

(Cut 2)



CROTCH
REINFORCEMENT PIECE

(Cut 1)

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